To map out Alex's coffee-ordering journey, step back from drawing frames and focus on extracting data into a systematic grid. Journey mapping shifts the focus from visual storytelling to analyzing UX touchpoints, user psychology, and software opportunities across sequential phases.

1.1. Define Scope & Persona: Establish boundaries before gathering data.

- **Persona:** Alex (Time-crunched tech worker on a daily train commute).
- **Scenario:** Pre-ordering coffee to skip the line before arriving at the office.

Goal: Understand where friction occurs in the ordering/pickup pipeline to identify app features and UX improvements.

2.2. Map the High-Level Experience Timeline: Establish the horizontal axis.
Break the user's end-to-end task into distinct chronological phases:

- **Discovery / Trigger:** Realizing a need or receiving a prompt on the train.
- **Selection:** Browsing the menu and customizing an item.
- **Checkout:** Reviewing order details and submitting payment.
- **Fulfillment:** Arriving at the store, locating the item, and picking it up.
- **Post-Purchase:** Consuming the coffee and reflecting on the experience.

3.3. Define Your Data Rows: Establish the vertical axis.
Build a structured table template with standard UX tracking criteria:

User Actions: What physical or digital steps is Alex taking?
Touchpoints: What interface or physical element is being interacted with
Thoughts: What is Alex thinking during this exact step?
Emotional State: Is frustration high or low? (often visually tracked as a curve).
Pain Points: Where does friction, uncertainty, or delay occur?
Opportunities: What software capabilities or UI tweaks solve those pain points?

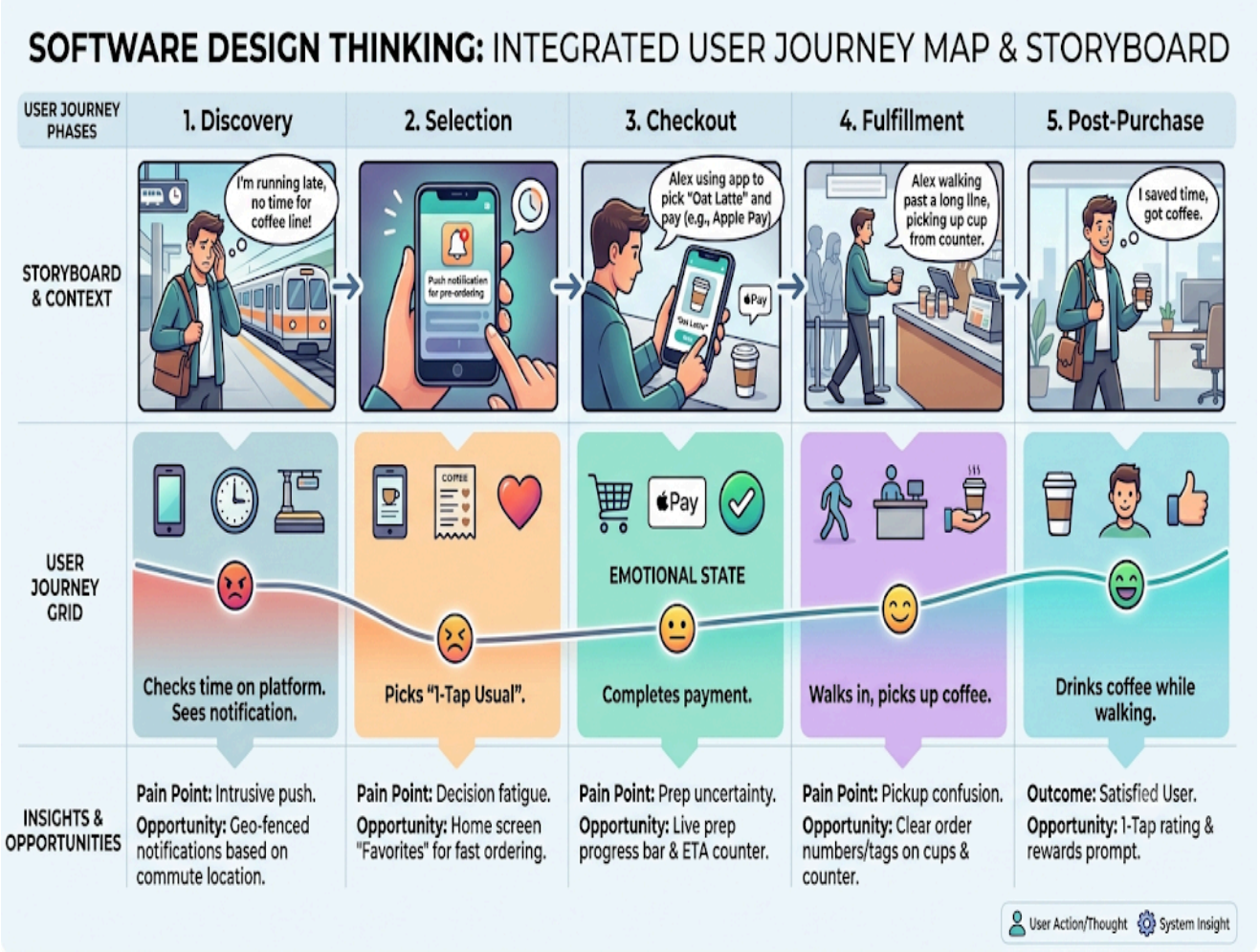
4.4. Plot the Journey Data & Identify Opportunities:
Populate and analyze grid cells.
Fill in each cell using real user research, analytics, or team ideation.

- Look specifically for **dips in emotion** (e.g., Alex worrying if the coffee will actually be warm upon arrival).

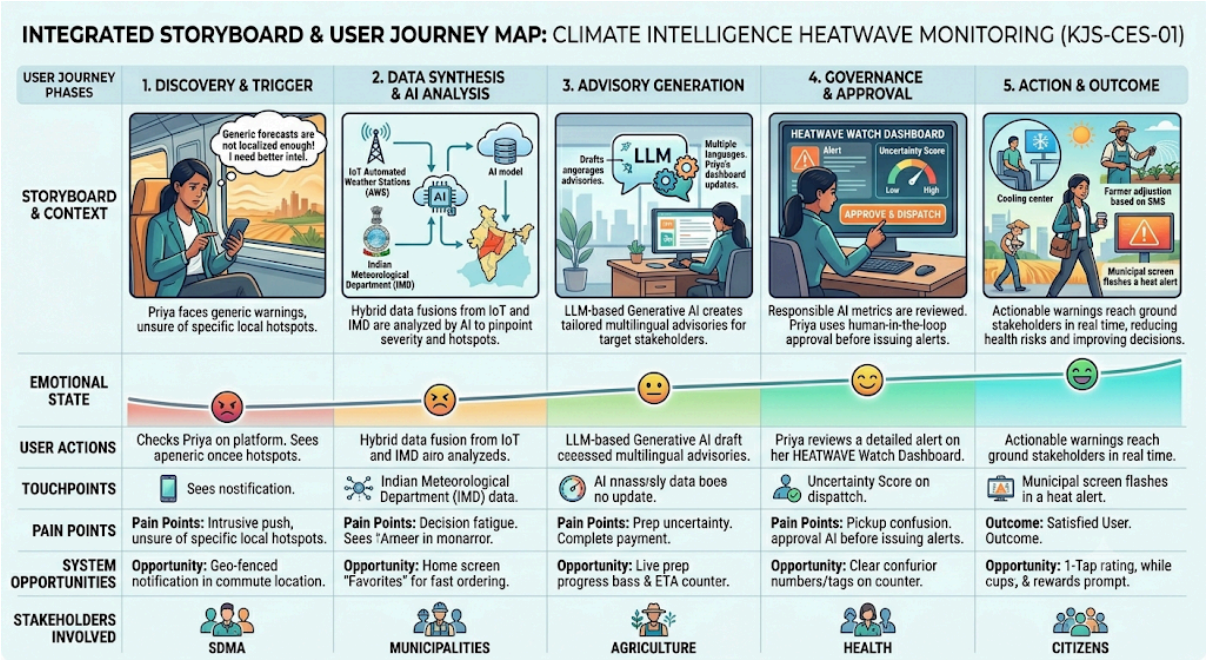
- Translate each pain point directly into a candidate for your software product backlog (e.g., "Add real-time prep status timer").

Best Practices for User Journey Mapping:

- **Base it on research, not assumptions:** Gather real user feedback or observe actual customer behavior rather than relying entirely on internal team guesses.
- **Focus on the handoffs:** Pay close attention to points where the user transitions between digital interfaces and real-world environments (e.g., moving from the app checkout screen to walking into the physical store).
- **Involve cross-functional teams:** Build the map collaboratively with product managers, UX designers, software engineers, and customer support leads so everyone gains a shared understanding of user pain points.



Heatwave Monitoring



Agriculture:Sugarcane

Based on the comprehensive project framework provided for the AI-Powered Irrigation Advisory System for Sugarcane (Use Case ID: KJS-AGR-01), I have designed and drawn the requested **Storyboard** and **User Journey Map**.

This design captures the holistic flow—from technical data acquisition by engineering teams to the actual interaction with the farmer on the ground—incorporating the modern AI governance and agentic paradigms defined in the document.

Part 1: Storyboard (Visual Narrative & Context)

The storyboard visualizes the lifecycle of the advisory system, illustrating how data from spatially mapped farms flows through AI systems to create a simple, action-oriented recommendation for the end beneficiary.

1.Scene 1: The Intuition Problem (KJS-AGR-01): Trigger / Current State.

- **Visual:** Farmer Appa Saheb in his digitized Karnataka sugarcane plot. He looks confused, touching dry soil. Generic weather forecast says "Partly Cloudy."
- **Thought Bubble:** *"Generic forecasts aren't enough. Should I over-irrigate to be safe, or wait? Water is wasting."*
- **Caption:** Decisions are manual, experience-based, ignoring scientific soil moisture or micro-climatic data[cite: 1].

2.Scene 2: Sensor and Data Acquisition: Technical Execution (Part B/C/D).

- **Visual:** Multi-disciplinary engineering teams deploying hardware. One student (EXTC/ECE) installs an IoT gateway[cite: 1]; another (RAI/ETRX) calibrates soil moisture sensors. Satellite icons represent GIS mapping[cite: 1].
- **Dialogue:** student: *"Sensors transmitting. Calibrating connection."*
- **Caption:** KJSSE and KJSIT engineering departments collaborate to deploy IoT sensors and transmit plot-level agricultural data[cite: 1].

3.Scene 3: AI Engine & Advisory Generation (Human-in-the-Loop):AI Analytics (Stage 3/4/6).

- **Visual:** An operations center. Data pipelines flow through an AI model block showing "Yield Stress Probability" and "Irrigation Prediction." An Agronomist (Human-in-the-Loop[cite: 1]) verifies an alert.
- **Dialogue:** system: *"Plot K-12: Stress imminent. LLM converting prediction to advisory."* agronomist: *"Forecast verified."*
- **Caption:** AI/ML models analyze sensor and weather data[cite: 1]. A Generative AI layer (LLM[cite: 1]) converts technical outputs into multilingual advisories (e.g., Kannada/Marathi)[cite: 1].
- **System (Agentic AI):** Proactive monitoring identifies stress probability.

4.Scene 4: The Advisory (Personalized Precision): User Interface / UI (Stage 7/9/image_4.png).

- **Visual:** Appa Saheb receives a simple, bold notification on his phone app. The screen is a simplified view of image_4.png.
- **Notification Text:** ***"Irrigate Plot K-12 (Sugarcane) Tomorrow Morning for 2.5 Hours."***
- **Caption:** Delivery of personalized recommendations via farmer-friendly mobile platforms for precision irrigation scheduling[cite: 1].
- **Responsible AI:** Explainable advisory is simple and tailored to plot conditions.

5.Scene 5: Implementation & Resilient Future: Impact / Outcome (Objectives).

- **Visual:** Appa Saheb activates his pump precisely according to the advisory. He looks relieved. A final split screen shows: reduced water wastage, optimized electricity usage, and healthy sugarcane growth with nutrient losses avoided[cite: 1].
- **Thought Bubble:** *"Data-driven irrigation. Water saved. Crop healthy.*
- **Caption:** Expected outputs: Enhanced water use efficiency, reduced health risks for crops, and climate resilience[cite: 1].

Part 2: User Journey Map (KJS-AGR-01)

This journey map structures the same scenario chronologically, identifying exactly what the user (Farmer Appa Saheb) is doing, thinking, and feeling, and mapping these to the specific system operations and inter-disciplinary roles.

Journey Phase	**1. Threat Detection**	**2. Plot Diagnosis**	**3. Alert Review & Action**	**4. Field Implementation**	**5. Success Tracking**
Storyboard Scene	Scene 1 (Trigger)	Scene 3 (AI Engine)	Scene 4 (UI Delivery)	Scene 5 (Impact)	Scene 5 (Impact)
User Action (Farmer Appa Saheb)	Observes crop/weather. Wonders if irrigation is needed	*(System analyzes background)*. Opens mobile dashboard	Receives & reads prioritized "Next Irrigation" notification	Follows precise "Date, Time, Duration" recommendation	Reviews "Yield Prediction" & soil history on app
System Phase	Field Data Acquisition (Stage 1	AI Analytics & Advisory Generation (Stage 4/6)	Dashboard Delivery & Farmer Interaction (Stage 7/9)	Smart Irrigation Automation (Stage 8)	Continuous Learning & Feedback (Stage 10)

Journey Phase	1. Threat Detection	2. Plot Diagnosis	3. Alert Review & Action	4. Field Implementation	5. Success Tracking
User Thoughts	*"Intuition tells me yes, but is it over-irrigating?"*	*"Where is the scientific recommendation?"*	*"It says tomorrow, 2.5 hours. Simple."*	*"Pump scheduled. Water productivity improved."*	*"Crop water requirement was met; yield stress avoided."*
Emotional State	😟 Anxious / Uncertainty	😐 Focused	📈 Relieved / Informed	😊 Confident / Empowered	😊 Satisfied / Resilient
System Operations	Data integration from IMD/IoT/GIS (KJSSE).	Predicting stress probability, date, duration.	Conversion of technical outputs (CE) to LLM advice (IT/AI&DS).	(Optional) Automated pump scheduling (ECE).	Feedback loop on advisory acceptance/override.
Pain Points	Water wastage, nutrient losses, high cultivation cost.	Missing/noisy sensor data or limited labeled datasets].	Difficulty monitoring thousands of farms.	Village connectivity issues / human mistrust.	Ensuring explainable recommendations to maintain trust.
Opportunities	Plot-level monitoring using sensor networks and geofencing	Multilingual advisor generated for target stakeholder segments.	Personalized proactive decision-making via	Plot-specific irrigation automation (Stage 8) / operational optimization.	Continuous monitoring and adaptive self-learning systems (RAI.

Journey Phase	1. Threat Detection	2. Plot Diagnosis	3. Alert Review & Action	4. Field Implementation	5. Success Tracking
	.		Agentic workflows .		